

Arduino-Controlled 12V BATTERY CHARGER

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The circuit presented here can automatically charge a 12V, 7Ah battery, or above. Special features of the charger are as follows. It automatically controls the charging current as per the status of the battery. Battery voltage level as well as charging status are indicated on the LCD display. The charger maintains float voltage, if battery is fully charged. Arduino identifies status of the battery connection and voltage,

and indicates the same on the LCD.

Circuit and working

The circuit diagram of the Arduino-controlled 12V battery charger is shown in Fig. 1. It is built around Arduino Uno (Board1), adjustable voltage regulator LM338 (IC1), 12V voltage regulator 7812 (IC2), 16x2 LCD (LCD1) and a few other components.

When AC mains power is on, step-down transformer X1 generates about

15V AC and converts it to above 18V DC after rectification and filtering. DC voltage thus available is split into two: one for charging the 12V battery (BATT.1) through IC1 and the other for power supply to Arduino (Board1) through IC2.

While charging the battery, its voltage is displayed on the left-top corner of LCD1. The charger/charging type and status is displayed on the second line on LCD1. LM338 supplies 14.2V

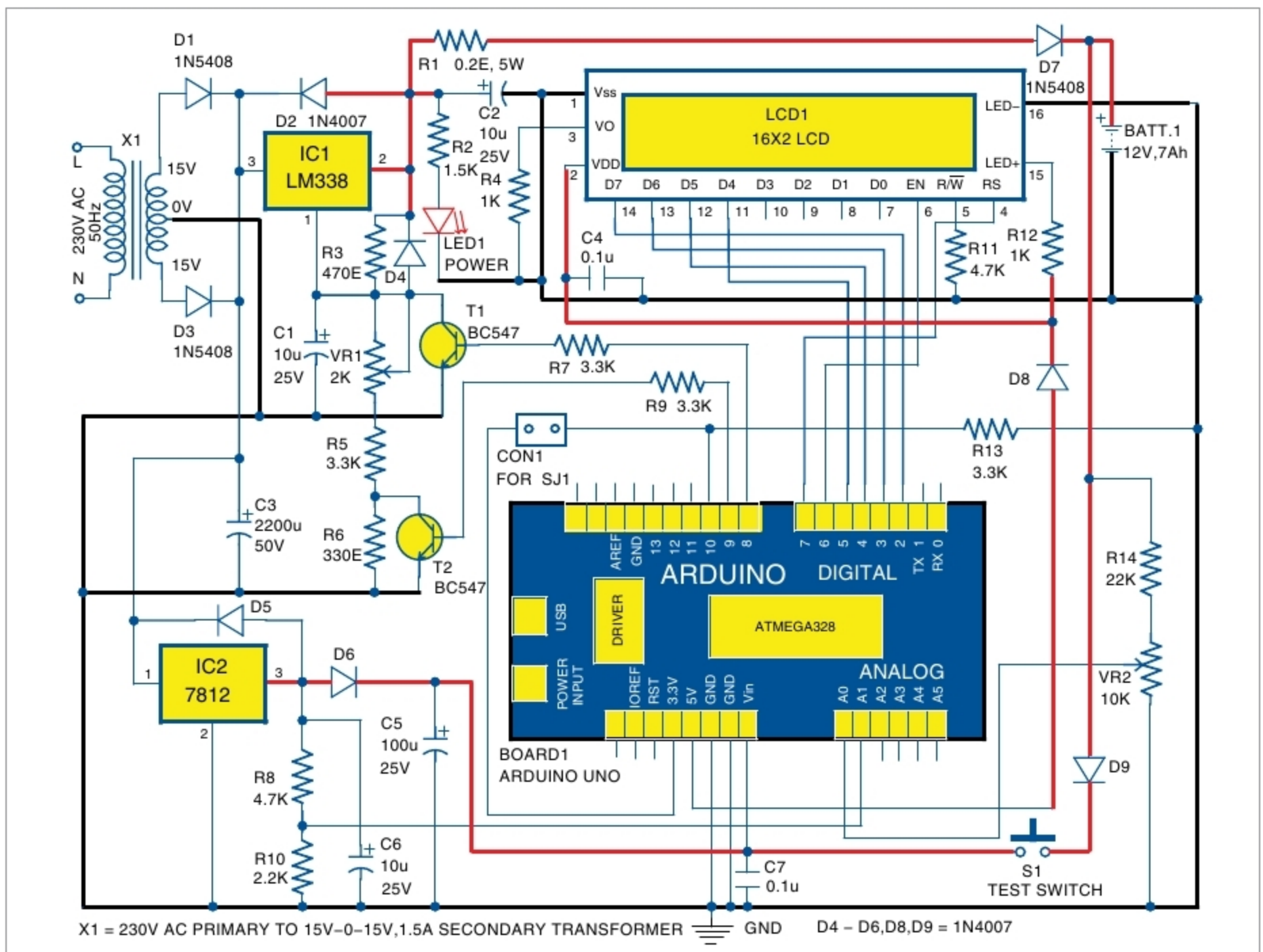


Fig. 1: Circuit diagram of battery charger